Guidance and Control of Unmanned Autonomous Vehicles

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UAV: Longitudinal State-Space Equations

Dynamics in longitudinal direction

$$
\begin{aligned}
\dot{u} =& rv - qw - g\sin\theta + \frac{\rho V_a^2 S}{2m}\left(C_{X_0} + C_{X_\alpha}\alpha + C_{X_q}\frac{cq}{2V_a} + C_{X_{\delta_e}}\delta_e\right) \\
&+ \frac{\rho S_{\text{prop}} C_{\text{prop}}}{2m}\left(k_{\text{motor}}^2\delta_t^2 - V_a^2\right) \\
\dot{w} =& qu - pv + g\cos\theta\cos\phi + \frac{\rho V_a^2 S}{2m}\left(C_{Z_0} + C_{Z_\alpha}\alpha + C_{Z_q}\frac{cq}{2V_a} + C_{Z_{\delta_e}}\delta_e\right) \\
\dot{q} =& \frac{J_{xz}}{J_y}(r^2 - p^2) + \frac{J_z - J_x}{J_y}pr + \frac{1}{2J_y}\rho V_a^2 cS\left[C_{m_0} + C_{m_\alpha}\alpha + C_{m_q}\frac{cq}{2V_a} + C_{m_{\delta_e}}\delta_e\right] \\
\dot{\theta} =& q\cos\phi - r\sin\phi \\
\dot{h} =& u\sin\theta - v\sin\phi\cos\theta - w\cos\phi\cos\theta
\end{aligned}
$$

- For zero lateral states: $\phi = p = r = \beta = v = 0$, $\alpha = \tan^{-1}\left(\frac{w}{u}\right)$; $V_a = \sqrt{u^2 + w^2}$

UAV: Longitudinal State-Space Equations

Dynamics in longitudinal direction

$$
\begin{aligned}
\dot{u} =& rv - qw - g\sin\theta + \frac{\rho S(u^2+w^2)}{2m}\left(C_{X_0} + C_{X_\alpha}\tan^{-1}\left(\frac{w}{u}\right) + C_{X_{\delta_e}}\delta_e\right) \\
&+ \frac{\rho S_{\text{prop}} C_{\text{prop}}}{2m}\left(k_{\text{motor}}^2\delta_t^2 - (u^2+w^2)\right) + \frac{\rho S\sqrt{u^2+w^2}}{4m}C_{X_q}cq \\
\dot{w} =& qu + g\cos\theta + \frac{\rho S(u^2+w^2)}{2m}\left(C_{Z_0} + C_{Z_\alpha}\tan^{-1}\left(\frac{w}{u}\right) + C_{Z_{\delta_e}}\delta_e\right) \\
&+ \frac{\rho S\sqrt{u^2+w^2}}{4m}C_{Z_q}cq \\
\dot{q} =& \frac{1}{2J_y}\rho c S(u^2+w^2)\left[C_{m_0} + C_{m_\alpha}\tan^{-1}\left(\frac{w}{u}\right) + C_{m_{\delta_e}}\delta_e\right] + \frac{\rho S\sqrt{u^2+w^2}}{4J_y}C_{m_q}c^2q \\
\dot{\theta} =& q\cos\phi - r\sin\phi \\
\dot{h} =& u\sin\theta - v\sin\phi\cos\theta - w\cos\phi\cos\theta
\end{aligned}
$$

UAV: Longitudinal State-Space Equations

- Linearized equations

$$\boxed{\dot{\bar{\boldsymbol{x}}} \approx \frac{\partial \boldsymbol{f}(\boldsymbol{x}^*, \boldsymbol{u}^*)}{\partial \boldsymbol{x}} \bar{\boldsymbol{x}} + \frac{\partial \boldsymbol{f}(\boldsymbol{x}^*, \boldsymbol{u}^*)}{\partial \boldsymbol{u}} \bar{\boldsymbol{u}}}$$

- Jacobians of equations

$$\frac{\partial \boldsymbol{f}_{\text{long}}}{\partial \boldsymbol{x}_{\text{long}}} = \begin{pmatrix} \frac{\partial \dot{u}}{\partial u} & \frac{\partial \dot{u}}{\partial w} & \frac{\partial \dot{u}}{\partial q} & \frac{\partial \dot{u}}{\partial \theta} & \frac{\partial \dot{u}}{\partial h} \\ \frac{\partial \dot{w}}{\partial u} & \frac{\partial \dot{w}}{\partial w} & \frac{\partial \dot{w}}{\partial q} & \frac{\partial \dot{w}}{\partial \theta} & \frac{\partial \dot{w}}{\partial h} \\ \frac{\partial \dot{q}}{\partial u} & \frac{\partial \dot{q}}{\partial w} & \frac{\partial \dot{q}}{\partial q} & \frac{\partial \dot{q}}{\partial \theta} & \frac{\partial \dot{q}}{\partial h} \\ \frac{\partial \dot{\theta}}{\partial u} & \frac{\partial \dot{\theta}}{\partial w} & \frac{\partial \dot{\theta}}{\partial q} & \frac{\partial \dot{\theta}}{\partial \theta} & \frac{\partial \dot{\theta}}{\partial h} \\ \frac{\partial \dot{h}}{\partial u} & \frac{\partial \dot{h}}{\partial w} & \frac{\partial \dot{h}}{\partial q} & \frac{\partial \dot{h}}{\partial \theta} & \frac{\partial \dot{h}}{\partial h} \end{pmatrix}, \quad \frac{\partial \boldsymbol{f}_{\text{long}}}{\partial \boldsymbol{u}_{\text{long}}} = \begin{pmatrix} \frac{\partial \dot{u}}{\partial \delta_e} & \frac{\partial \dot{u}}{\partial \delta_t} \\ \frac{\partial \dot{w}}{\partial \delta_e} & \frac{\partial \dot{w}}{\partial \delta_t} \\ \frac{\partial \dot{q}}{\partial \delta_e} & \frac{\partial \dot{q}}{\partial \delta_t} \\ \frac{\partial \dot{\theta}}{\partial \delta_e} & \frac{\partial \dot{\theta}}{\partial \delta_t} \\ \frac{\partial \dot{h}}{\partial \delta_e} & \frac{\partial \dot{h}}{\partial \delta_t} \end{pmatrix}$$

UAV: Longitudinal State-Space Equations

Linearized equations in the longitudinal direction

$$\begin{bmatrix} \dot{\bar{u}} \\ \dot{\bar{w}} \\ \dot{\bar{q}} \\ \dot{\bar{\theta}} \\ \dot{\bar{h}} \end{bmatrix} = \begin{bmatrix} X_u & X_w & X_q & -g\cos\theta^* & 0 \\ Z_u & Z_w & Z_q & -g\sin\theta^* & 0 \\ M_u & M_w & M_q & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ \sin\theta^* & -\cos\theta^* & 0 & u^*\cos\theta^* + w^*\sin\theta^* & 0 \end{bmatrix} \begin{bmatrix} \bar{u} \\ \bar{w} \\ \bar{q} \\ \bar{\theta} \\ \bar{h} \end{bmatrix} + \begin{bmatrix} X_{\delta_e} & X_{\delta_t} \\ Z_{\delta_e} & 0 \\ M_{\delta_e} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \bar{\delta}_e \\ \bar{\delta}_t \end{bmatrix}$$

- Can we use an angle of attack to express the longitudinal dynamics? Yes, How?
- Use of angle of attack α in place of $\bar{w}$, with zero sideslip angle $\beta = 0$,

$$w = V_a \sin\alpha\cos\beta = V_a \sin\alpha \implies \bar{w} = V_a \cos\alpha^* \bar{\alpha} \implies \boxed{\dot{\bar{\alpha}} = \left(\frac{1}{V_a^* \cos\alpha^*}\right)\dot{\bar{w}}}$$

UAV: Longitudinal State-Space Equations

Linearized equations in the longitudinal direction

$$\begin{bmatrix} \dot{\bar{u}} \\ \dot{\bar{\alpha}} \\ \dot{\bar{q}} \\ \dot{\bar{\theta}} \\ \dot{\bar{h}} \end{bmatrix} = \begin{bmatrix} X_u & X_w V_a^* \cos\alpha^* & X_q & -g\cos\theta^* & 0 \\ \dfrac{Z_u}{V_a^* \cos\alpha^*} & Z_w & \dfrac{Z_q}{V_a^* \cos\alpha^*} & -\dfrac{g\sin\theta^*}{V_a^* \cos\alpha^*} & 0 \\ M_u & M_w V_a^* \cos\alpha^* & M_q & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ \sin\theta^* & -V_a^* \cos\alpha^* \cos\theta^* & 0 & u^* \cos\theta^* + w^* \sin\theta^* & 0 \end{bmatrix} \begin{bmatrix} \bar{u} \\ \bar{\alpha} \\ \bar{q} \\ \bar{\theta} \\ \bar{h} \end{bmatrix}$$

$$+ \begin{bmatrix} X_{\delta_e} & X_{\delta_t} \\ \dfrac{Z_{\delta_e}}{V_a^* \alpha^*} & 0 \\ M_{\delta_e} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \bar{\delta}_e \\ \bar{\delta}_t \end{bmatrix}$$

UAV: Modes

- What are different modes of the vehicle's motion?
- Aircraft modes
  - **Longitudinal**: Short-period (SP) and Phugoid
  - **Lateral**: Rolling, Spiral divergence, and Dutch-roll (DR)

For constant altitude and throttle inputs

$$\begin{bmatrix} \dot{\bar{u}} \\ \dot{\bar{\alpha}} \\ \dot{\bar{q}} \\ \dot{\bar{\theta}} \end{bmatrix} = \begin{bmatrix} X_u & X_w V_a^* \cos\alpha^* & X_q & -g\cos\theta^* \\ \frac{Z_u}{V_a^* \cos\alpha^*} & Z_w & \frac{Z_q}{V_a^* \cos\alpha^*} & -\frac{g\sin\theta^*}{V_a^* \cos\alpha^*} \\ M_u & M_w V_a^* \cos\alpha^* & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \bar{u} \\ \bar{\alpha} \\ \bar{q} \\ \bar{\theta} \end{bmatrix} + \begin{bmatrix} X_{\delta_e} \\ \frac{Z_{\delta_e}}{V_a^* \cos\alpha^*} \\ M_{\delta_e} \\ 0 \end{bmatrix} \bar{\delta}_e$$

- How to identify different modes of vehicles?
- **Eigenvalues**: $\underbrace{\text{One fast and heavily damped}}_{\text{Short}-\text{Period}}$; $\underbrace{\text{One slow and lightly damped}}_{\text{Phugoid}}$

Longitudinal Modes

- System matrix with state vector as $[\bar{u}\ \bar{w}\ \bar{q}\ \bar{\theta}]^\top$

$$A = \begin{bmatrix} -0.006868 & 0.01395 & 0 & -32.2 \\ -0.09055 & -0.3151 & 773.98 & 0 \\ 0.0001187 & -0.001026 & -0.4285 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

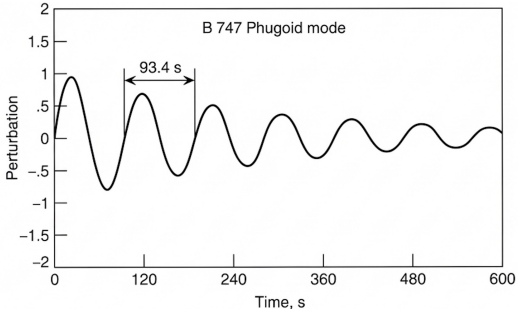
- Characteristic equation

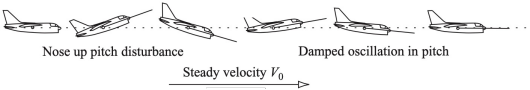
$$\lambda^4 + 0.750468\lambda^3 + 0.935494\lambda^2 + 0.0094630\lambda + 0.0041959 = 0$$

- Eigenvalues: $\lambda_{1,2} = -0.003289 \pm 0.06723i,\ \lambda_{3,4} = -0.3719 \pm 0.8875i$
- Which eigenvalue corresponds to which mode and why?

$$\underbrace{\lambda_{1,2} = -0.003289 \pm 0.06723i}_{\text{Phugoid mode}},\ \underbrace{\lambda_{3,4} = -0.3719 \pm 0.8875i}_{\text{Short-period mode}}$$

Longitudinal Modes





Mode	Name	Period	t_{half}	N_{half}
1	Phugoid	93.4	211	22.5
2	Short-period	7.08	1.86	0.26

Longitudinal Modes: Eigenvectors

	Phugoid		Short-Period	
	Magnitude	Phase	Magnitude	Phase
$\Delta\hat{u}$	0.62	92.4°	0.029	57.4°
$\alpha = \hat{w}$	0.036	82.8°	1.08	19.2°
$\hat{q}$	0.0012	92.8°	0.017	112.7°
$\Delta\theta$	1.0	0°	1.0	0°

- Normalized eigenvectors to see the relative magnitude.
- **Phugoid**: Small changes in $\bar{q}$ and $\bar{\alpha}$, but $\bar{u}$ and $\bar{\theta}$ are present with significant magnitude. Speed $\bar{u}$ leads $\bar{\theta}$ leads by 90° in phase.
- **Short-period**: Negligible speed variation, while $\bar{\alpha}$ oscillates with an amplitude and phase similar to $\bar{\theta}$. Two degrees of freedom, $\bar{\theta}$ and α.

UAV: Short period mode

- For Short-period mode, u is almost constant, that is, $\bar{u} = \dot{\bar{u}} = 0$

$$\boxed{\dot{\bar{\alpha}} = Z_w \bar{\alpha} + \frac{Z_q}{V_a^* \cos\alpha}\dot{\bar{\theta}} - \frac{g\sin\theta^*}{V_a^* \cos\alpha^*}\bar{\theta} + \frac{Z_{\delta_e}}{V_a^* \cos\alpha^*}\bar{\delta}_e} \quad \boxed{\ddot{\bar{\theta}} = M_w V_a^* \cos\alpha^* \bar{\alpha} + M_q \dot{\bar{\theta}}}$$

$$\begin{bmatrix} s - Z_w & -\frac{Z_q s}{V_a^* \cos\alpha} + \frac{g\sin\theta^*}{V_a^* \cos\alpha^*} \\ -M_w V_a^* \cos\alpha^* & s(s - M_q) \end{bmatrix} \begin{bmatrix} \bar{\alpha}(s) \\ \bar{\theta}(s) \end{bmatrix} = \begin{bmatrix} \frac{Z_{\delta_e}}{V_a^* \cos\alpha^*} \\ 0 \end{bmatrix} \bar{\delta}_e(s)$$

- For $\theta^* = 0$, the characteristic equation

$$s\left\{s^2 - (Z_w + M_q)s + M_q Z_w - M_w Z_q\right\} = 0$$

- **Short period poles**:

$$\lambda_{\text{short}} = \frac{Z_w + M_q}{2} \pm \sqrt{\left(\frac{Z_w + M_q}{2}\right)^2 - M_q Z_w + M_w Z_q}$$

UAV: Phugoid mode

For constant angle of attack, $\alpha = \alpha^*$, we have $\bar{\alpha} = \dot{\bar{\alpha}} = 0$,

$$\begin{bmatrix} \dot{\bar{u}} \\ 0 \\ \dot{\bar{q}} \\ \dot{\bar{\theta}} \end{bmatrix} = \begin{bmatrix} X_u & X_w V_a^* \cos\alpha^* & X_q & -g\cos\theta^* \\ \frac{Z_u}{V_a^* \cos\alpha^*} & Z_w & \frac{Z_q}{V_a^* \cos\alpha^*} & -\frac{g\sin\theta^*}{V_a^* \cos\alpha^*} \\ M_u & M_w V_a^* \cos\alpha^* & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \bar{u} \\ 0 \\ \bar{q} \\ \bar{\theta} \end{bmatrix} + \begin{bmatrix} X_{\delta_e} \\ \frac{Z_{\delta_e}}{V_a^* \cos\alpha^*} \\ M_{\delta_e} \\ 0 \end{bmatrix} \bar{\delta}_e$$

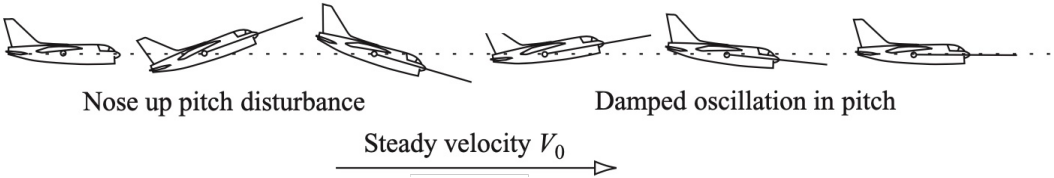
- On taking the Laplace transform of the first two equations

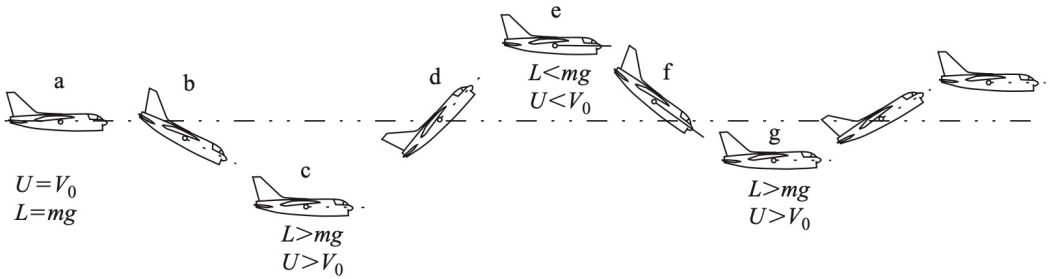
$$\begin{bmatrix} s - X_u & -X_q s + g\cos\theta^* \\ -Z_u & -Z_q s + g\sin\theta \end{bmatrix} \begin{bmatrix} \bar{u}(s) \\ \bar{\theta}(s) \end{bmatrix} = \begin{bmatrix} X_{\delta_e} \\ Z_{\delta_e} \end{bmatrix} \bar{\delta}_e(s)$$

- For $\theta^* = 0$, the characteristic equation

$$s^2 + \left(\frac{Z_u X_q - X_u Z_q}{Z_q}\right) s - \frac{g Z_u}{Z_q} = 0 \implies \lambda = -\frac{Z_u X_q - X_u Z_q}{2 Z_q} \pm \sqrt{\left(\frac{Z_u X_q - X_u Z_q}{Z_q}\right)^2 + \frac{g Z_u}{Z_q}}$$

UAV: Longitudinal modes (Shirt-Period and Phugoid mode) Behavior





UAV: Lateral modes

For $\bar{\theta} = 0$, the lateral dynamics

$$\begin{bmatrix} \dot{\bar{\beta}} \\ \dot{\bar{p}} \\ \dot{\bar{r}} \\ \dot{\bar{\phi}} \end{bmatrix} = \begin{bmatrix} Y_v & \frac{Y_p}{V_a^* \cos\beta^*} & \frac{Y_r}{V_a^* \cos\beta^*} & \frac{g\cos\theta^* \cos\phi^*}{V_a^* \cos\beta^*} \\ L_v V_a^* \cos\beta^* & L_p & Y_r & 0 \\ N_v V_a^* \cos\beta^* & N_p & Y_r & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \bar{\beta} \\ \bar{p} \\ \bar{r} \\ \bar{\phi} \end{bmatrix} + \begin{bmatrix} \frac{Y_{\delta_a}}{V_a^* \cos\beta^*} & \frac{Y_{\delta_r}}{V_a^* \cos\beta^*} \\ Y_{\delta_a} & Y_{\delta_r} \\ Y_{\delta_a} & Y_{\delta_r} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \bar{\delta}_a \\ \bar{\delta}_r \end{bmatrix}$$

System matrix with the state vector as $[\bar{v}\ \bar{p}\ \bar{r}\ \bar{\phi}]^\top$

$$A = \begin{bmatrix} -0.0558 & 0 & -774 & 32.2 \\ -0.003865 & -0.4342 & 0.4136 & 0 \\ 0.001086 & -0.006112 & -0.1458 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Lateral Modes

- Eigenvalues

$$\lambda_1 = -0.0072973,\ \lambda_2 = -0.56248,\ \lambda_{3,4} = -0.033011 \pm 0.94655i$$

- What can you say about the lateral modes?

$$\underbrace{\lambda_1 = -0.0072973}_{\text{Spiral mode}},\ \underbrace{\lambda_2 = -0.56248}_{\text{Roll mode}},\ \underbrace{\lambda_{3,4} = -0.033011 \pm 0.94655i}_{\text{Dutch Roll mode}}$$

- Two modes are convergences, one very rapid, one very slow
- One lightly damped oscillation with a period similar to that of the SP mode

Mode	Name	Period	t_{half}	N_{half}
1	Spiral		95	
2	Roll		1.23	
3	Dutch Roll	6.64	21	3.16

Lateral Modes: Spiral Mode

	Spiral		Rolling convergence		Dutch Roll	
	Magnitude	Phase	Magnitude	Phase	Magnitude	Phase
$\beta = v$	1.119×10^{-3}	$180°$	0.0198	$180°$	0.33	$-28.1°$
$\hat{p}$	1.63×10^{-4}	$0°$	0.0712	$180°$	0.12	$92.0°$
$\hat{r}$	9.20×10^{-4}	$180°$	0.0040	$0°$	0.037	$-112.3°$
ϕ	0.177	$180°$	1.0	$0°$	1.0	$0°$
ψ	1.0	$0°$	0.0562	$180°$	0.31	$155.7°$
$\frac{y_E}{u_0 t^*}$	7.772×10^3	$180°$	7.65	$0°$	1.69	$-165.8°$

- For spiral mode, $\boxed{\bar{\beta} : \bar{\phi} : \bar{\psi} = -0.001119 : -0.177 : 1}$
- Yawing at nearly **zero sideslip** with some rolling
- Aerodynamic variables $\boxed{\bar{\beta} : \bar{p} : \bar{r} = 1 : -0.137 : 0.773}$
- Largest of these (β) is negligibly small for the moderate values of $\bar{\phi}, \bar{\psi}$
- A weak mode because aerodynamic forces are very small.

Uncontrolled Motion: Roll and DR Modes

- For roll mode,

$$\boxed{\bar{\beta} : \bar{\phi} : \bar{\psi} = -0.0198 : 1 : -0.0625}$$

- Pure rotation around the x-axis
- Aerodynamic variables

$$\boxed{\bar{\beta} : \bar{p} : \bar{r} = 0.278 : 1 : -0.0561}$$

- Largest rolling moment $C_{l_p}\bar{p}$, $\bar{r}$ contribution is small.
- **In DR mode, $\bar{\beta}, \bar{\phi}, \bar{\psi}$ are of same magnitude**.
- $\bar{r}$ is one order smaller while $\bar{\beta}$ and $\bar{\psi}$ are almost equal and opposite.
- Rectilinear motion with yawing and rolling, with rolling lagging yawing by around 160°

UAV: Roll mode

- Dynamics of roll rate $\bar{p}$

$$\dot{\bar{p}} = L_v V_a^* \cos\beta \bar{\beta} + L_p \bar{p} + L_r \bar{r} + L_{\delta_a} \bar{\delta}_a + L_{\delta_r} \bar{\delta}_r$$

- Rolling mode: Using $\bar{\beta} = \bar{r} = \bar{\delta}_r = 0$

$$\dot{\bar{p}} = L_p \bar{p} + L_{\delta_a} \bar{\delta}_a$$

- Transfer function

$$\frac{\bar{p}(s)}{\bar{\delta}_a(s)} = \frac{L_{\delta_a}}{s - L_p}$$

- Eigenvalue of rolling mode

$$\lambda_{\text{rolling}} = L_p$$

UAV: Spiral-divergence mode

- For Spiral-divergence mode, assume $\dot{\bar{p}} = \bar{p} = 0$, and $\bar{\delta}_r \approx 0$

$$\boxed{0 = L_v V_a^* \cos\beta^* \bar{\beta} + L_r \bar{r} + L_{\delta_a} \bar{\delta}_a} \qquad \boxed{\dot{\bar{r}} = N_v V_a^* \cos\beta^* \bar{\beta} + N_r \bar{r} + N_{\delta_a} \bar{\delta}_a}$$

- On solving after eliminating β,

$$\dot{\bar{r}} = \left(\frac{N_r L_v - N_v L_r}{L_v}\right) \bar{r} + \left(\frac{N_{\delta_a} L_v - N_v L_{\delta_a}}{L_v}\right) \bar{\delta}_a$$

$$\frac{\bar{r}(s)}{\bar{\delta}_a(s)} = \frac{(N_{\delta_a} L_v - N_v L_{\delta_a})/L_v}{s - (N_r L_v - N_v L_r)/L_v}$$

- **Pole of spiral mode**: Typically in RH plane and unstable mode

$$\lambda_{\text{spiral}} = \frac{N_r L_v - N_v L_r}{L_v}$$

UAV: Dutch-Roll mode

- On neglecting the rolling motion of aircraft

$$\begin{bmatrix} \dot{\bar{\beta}} \\ \dot{\bar{r}} \end{bmatrix} = \begin{bmatrix} Y_v & \dfrac{Y_r}{V_a \cos\beta^*} \\ N_v V_a \cos\beta^* & N_r \end{bmatrix} \begin{bmatrix} \bar{\beta} \\ \bar{r} \end{bmatrix} + \begin{bmatrix} \dfrac{Y_{\delta_r}}{V_a \cos\beta^*} \\ N_{\delta_r} \end{bmatrix} \bar{\delta}_r$$

- Characteristic equation: $s^2 - (Y_v + N_r)s + (Y_v N_r - N_v Y_r) = 0$

$$\lambda_{\text{Dutch-Roll}} = \frac{Y_v + N_r}{2} \pm \sqrt{\left(\frac{Y_v + N_r}{2}\right)^2 - (Y_v N_r - N_v Y_r)}$$

- Transfer function from rudder deflection δ_r to the yaw rate r

$$\frac{r(s)}{\delta_r(s)} = \begin{bmatrix} 0 & 1 \end{bmatrix} \left[s\boldsymbol{I} - \begin{bmatrix} Y_v & \frac{Y_r}{V_a^* \cos\beta^*} \\ N_v V_a^* \cos\beta^* & N_r \end{bmatrix} \right]^{-1} \begin{bmatrix} \frac{Y_{\delta_r}}{V_a^* \cos\beta^*} \\ N_{\delta_r} \end{bmatrix}$$

$$= \frac{N_{\delta_r} s + (Y_{\delta_r} N_v - N_{\delta_r} Y_v)}{s^2 - (Y_v + N_r)s + (Y_v N_r - Y_r N_v)}$$

Text-Books/References

References

- Randal Beard and Timothy W. McLain, "Small Unmanned Aircraft: Theory and Practice", Princeton University Press, 2012
- Bernard Etkin and Lloyd Duff Reid, "Dynamics of Flight Stability and Control", John Wiley and Sons, Third Edition, 1996.

Thank you for your attention !!!